

Day 9 — Project Session 2: Cross-Cultural Comparison and Refinement

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a presentation-ready *rāśi*-wheel including a cross-cultural comparison element (Greek / Babylonian / Gregorian) and a rehearsed 3-minute presentation script.

Materials

- The in-progress *rāśi*-wheel from Day 8
- Reference printout: Sanskrit / Greek / Babylonian zodiac names side by side (provided below in *Cross-cultural reference*)
- Stopwatch / phone timer
- The rubric ([assessments/rubric.md](#))

Cross-cultural reference

The 12 zodiac signs across three traditions:

#	Sanskrit (Indian, <i>rāśi</i>)	Greek / English	Babylonian (~700 BCE origins)
1	<i>Meṣa</i> (Ram)	Aries (Ram)	LU.ḪUN.GÁ (Hired Man)
2	<i>Vṛṣabha</i> (Bull)	Taurus (Bull)	GU.AN.NA (Bull of Heaven)
3	<i>Mithuna</i> (Twins)	Gemini (Twins)	MAŠ.TAB.BA (Great Twins)
4	<i>Karka</i> (Crab)	Cancer (Crab)	AL.LUL (Crab)
5	<i>Siṃha</i> (Lion)	Leo (Lion)	UR.GU.LA (Lion)
6	<i>Kanyā</i> (Maiden)	Virgo (Maiden)	AB.SIN (Furrow / barley ear)
7	<i>Tulā</i> (Scales)	Libra (Scales)	RIN (Balance)
8	<i>Vṛścika</i> (Scorpion)	Scorpio (Scorpion)	GIR.TAB (Scorpion)
9	<i>Dhanus</i> (Bow)	Sagittarius (Archer)	PA.BIL.SAG (Archer)
10	<i>Makara</i> (Sea-goat)	Capricorn (Sea-goat)	SUḪUR.MAŠ (Goat-fish)
11	<i>Kumbha</i> (Water-pot)	Aquarius (Water-bearer)	GU.LA (Great One / water-pourer)
12	<i>Mīna</i> (Fishes)	Pisces (Fishes)	DU.NU.NU (Tails)

The Indian, Greek, and Babylonian names are *strikingly similar* — the same animals (ram, bull, twins, crab, lion, scorpion, fishes) appear in all three. This is not accidental. The most likely

historical explanation is **shared origin in Babylonian astronomy (~700 BCE)**, transmitted to Greece (post-Alexander, ~300 BCE) and to India (Hellenistic contact, last centuries BCE). The 12-fold ecliptic division and the imagery were absorbed into the Indian *jyotiṣa* tradition and merged with earlier Vedic star-lore (the 27 *nakṣatra* system, which is genuinely older and independently Indian).

The 27-*nakṣatra* system is distinctively Indian; the 12-*rāśi* system is the Babylonian-Greek-Indian common ground.

Lesson flow

Warm-up (5 min)

- Project the cross-cultural table on the board. Take 30 seconds to let students absorb.
- Prompt: “*What patterns do you see?*” — collect 2 observations.
- Likely observation: “they’re all about the same animals.” Confirm: yes — and this is a real historical clue about cultural exchange.

Core (5 min)

- One-paragraph summary on the board:
 - The 12-fold zodiac is a shared inheritance across Babylon, Greece, and India.
 - Indian *jyotiṣa* layered this on top of the older 27-*nakṣatra* system.
 - This means: when you study the *rāśis*, you are studying a 2700-year-old transcontinental mathematical-astronomical inheritance, not a purely Indian invention.
- This is **not** “Indians borrowed from the West” — Babylonian astronomy reached India before it reached Greece in some respects, and the Indian elaboration (the *nirayana* convention, the *ayanāṁśa* corrections) is a real intellectual contribution.

Refinement work (25 min)

Pairs refine their wheel:

- Add the cross-cultural element. Options:
 - A small inset showing the Babylonian name for ONE *rāśi* alongside the Sanskrit and Greek.
 - A single sentence on the wheel: “These 12 signs appear in similar form in Babylonian, Greek, and Indian astronomy — a 2700-year inheritance.”
 - A short comparative note on Cancer / Karka with a footnote on the Babylonian name.
- Finalise the written explanation paragraph (4–6 sentences) covering: axial tilt, the two *ayanas*, the four turning-points, the *sāyana* / *nirayana* distinction, and one citation.
- Decide who-says-what in the 3-minute presentation.

Rehearsal (8 min)

In groups of two pairs (4 students), pairs rehearse their 3-minute presentation in front of each other.

The listening pair times the speakers and gives one piece of feedback:

- ONE thing that came across clearly
- ONE thing the audience didn't understand

Then they swap.

Wrap-up (2 min)

- Exit signal: each pair raises hand showing 1, 2, 3, 4, or 5 fingers — how ready they feel for tomorrow.
- The teacher notes any pair showing 1 or 2 and plans a 5-minute conference at the start of Day 10.

Differentiation

- **One tier up:** Each pair preparing a “stretch” closer: a 30-second factoid the audience will remember (e.g. “Precession means in ~13,000 years, the December solstice will be in *Karka*, not *Makara*”). Confirm the factoid is astronomically accurate before they include it.
- **One tier down:** Skip the Babylonian table; just mention “the zodiac is shared between India, Greece, and the older Babylonian tradition.” That’s enough for a “Meets” presentation.

Teacher notes

- A small but real risk: a pair invents a Sanskrit etymology to “explain” a Greek name. Catch this — don’t let invented etymology into a presentation.
- The Babylonian names contain Akkadian / Sumerian cuneiform transliterations (LU.ḪUN.GÁ, etc.). Don’t drill these — they’re for visual comparison, not memorisation.
- Lateness on the wheel is acceptable as long as the presentation is rehearsed. Better an unfinished but well-explained wheel than a beautiful unstructured one.

Homework

- Final touches. By tomorrow morning, the wheel should be:
 - ☐ Labelled in Sanskrit and English / Greek
 - ☐ Coloured legibly
 - ☐ Showing all four turning-points with dual *sāyana* / *nirayana* dates
 - ☐ Including at least one cross-cultural reference
 - ☐ Accompanied by a 4–6 sentence explanation
 - ☐ Ready for a 3-minute presentation by both partners